

SHort-term Rhizosphere Response to Experimental Disturbance (ShRRED)

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Authors

Kayla C. Mathes, Tatum Delaney, Lisa Nalliah and Christopher M. Gough

Description

All code and data files in preparation for manuscript: *Dynamic rhizosphere processes stabilize soil respiration immediately following a phloem-disrupting disturbance.*

Location of study:

University of Michigan Biological Station, Pellston, MI

All updated analysis can be found in the script:

manuscript_script.R

R packages used:

library(googledrive)

library(dplyr)

library(plotrix)

library(car)

library(ggpubr)

library(gridExtra)

library(AICcmodavg)

library(agricolae)

library(gridExtra)

library(ggplot2)

library(rstatix)

library(cvequality)

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library(sjPlot)
library(sjmisc)
library(emmeans)
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Data sources and metadata:

Soil_respiration_ShRRED.csv

- Unique ID: unique identifier for each row
- Plot_ID: Plot within the experimental design (C1,C2,C3,C4,T1,T2,T3,T4)
- Type: Disturbance treatment (Treatment) or Control
- Plot_#: Plot nested within a treatment or control (1,2,3,4)
- Location: Soil collar location nested within a plot (1,2,3,4)
- Efflux: soil respiration measurement ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ sec}^{-1}$)
- Soil_VWC: Soil Volumetric water content to 20 cm (%)
- Soil_T: Soil temperature to 7cm (Celcius)
- Date: Date of measurement

CalculatedN_ShRRED_REU2022.csv

- Unique ID: unique identifier for each row
- Date_Collected: Date of field collection
- Treatment: Disturbance (Treatment) or Control
- Plot: Plot within the experimental design (C1,C2,C3,C4,T1,T2,T3,T4)
- Average_Soil_Temp:
- Wet_sample_weight_analysis: weight of soil (g) used in analysis for nitrogen content
- Wet_sample_weight_drying: weight of soil (g) to be used to determine gravimetric water content
- Dry_soil_weight: Dry soil weight (g) after drying for 24 hours
- Nitrate: concentration ($\mu\text{g/L}$)
- Ammonium: concentration ($\mu\text{g/L}$)
- grav_water_content: gravimetric water content (unitless)
- nitrate_calculated: Inorganic soil nitrate concentration mg NO_3^- per kg of dry soil
- ammonium_calculated: Inorganic soil nitrate concentration mg NH_4^+ per kg of dry soil

NSC_summary_mathes.csv

- Sample_Name: Unique identifier for tree
- Date: Date of field collection
- Core_Type: Root (R) or Bole (B) sample collection
- Treatment_Type: Disturbance (Treatment) or Control
- Plot_ID: Plot within the experimental design (C1,C2,C3,C4,T1,T2,T3,T4)
- Tree_type: Focal (F) or Neighboring (N)
- NSC: Non-structural carbohydrate concentrations (proportion % of NSC concentration to sample dry mass)

ShRRED_root_structure_6_26_22.csv

ShRRED_root_structure_7_13_22.csv

ShRRED_root_structure_7_25_22.csv

(selected columns used for analysis)

- Label: Unique ID
- Type: Disturbance (Treatment) or Control
- Plot: Plot within the experimental design (C1,C2,C3,C4,T1,T2,T3,T4)
- Dry_root_weight_g_measured: root weight
- Dry_root_adjusted_g: root weight after ash-free adjustments
- SoilVol.m3: Soil sample volume m3
- Length.cm: Total sample length (cm)
- ProjArea.cm2: Total area of sample (cm2)
- SurfArea.cm2: Surface area of sample (cm2)
- AvgDiam.mm: Average diameter of roots in sample (mm)
- RootVolume.cm3: Total root volume of sample (cm3)
- Tips: Number of root tips in the sample
- Forks: Number of root forks in the sample
- Crossings: number of crossings in sample

ShRRED_Stem_map_data.csv

- Plot_ID: Plot within the experimental design (C1,C2,C3,C4,T1,T2,T3,T4)
- Type: Disturbance (Treatment) or Control
- Plot_Haglof: Plot ID in Haglof Instrument
- Plot_radius: Plot radius from center (m)
- Tree_Nr: Unique tree number within each plot
- Tree_Spc: Tree species code:

- QURU: *Quercus rubra*
- PIST: *Pinus strobus*
- PIRE: *Pinus resinosa*
- ACRU: *Acer rubrum*
- POGR: *Populus grandidentata*
- FAGR: *Fagus grandifolia*
- Strata: Canopy position: Canopy (can) or Subcanopy (Sub)
- Tree_Dia: Diameter at breast Height (cm)
- Tree_PosTex1, Tree_PosTex2, Tree_PosTex3, Tree_Local_x, Tree_Local_y, Tree_Local_Dist, Tree_Local_Angle, Tree_Angle_ToPlotCenter: Haglof measurements to spatially locate tree within the plot
- Latitude
- Longitude